# Sayantan Mukhopadhyay

*Curriculum Vitae*

Senior Research Scholar (SRF)
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## *About Me*

I am a Senior Research Scholar at the National Institute of Science Education and Research (NISER), Bhubaneswar, working in the field of main-group organometallic chemistry. During my doctoral research under the guidance of Prof. Sharanappa Nembenna, I have developed broad expertise in the **design, synthesis and reactivity of metal complexes**, with a particular focus on **catalytic transformations and mechanistic investigations**. My research encompasses **organometallic synthesis, catalyst development, small-molecule activation, hydroboration, C–C bond formation and $CO_2$ activation**, supported by detailed kinetic, spectroscopic and structural studies. I am interested in **developing new organometallic systems, catalytic methodologies and mechanistic approaches**, and in exploring their applications across inorganic and organometallic chemistry.

## *Technical Expertise*

- *Air/moisture-sensitive reactions*
- *Glovebox Operation*
- *Schlenk line Operation*
- *Chromatography (TLC/CC)*
- *Crystallization*
- *Kinetic Study, Isotope labeling*
- *NMR / UV-VIS / IR Spectroscopy*
- *Mass Spectrometry*
- *Single crystal XRD*
- *OLEX 2.0 Software*
- *Endnote X9, Diamond 4.6 Software*
- *Chemdraw 16.0, Mast Renova 14.0*

## *Education*

**BSc Chemistry First Class (H)**
**2014–2017**
Vidyasagar *University*

**MSc Chemistry First Class**
**2017–2019**
*Chhatrapati Sahuji Maharaj University*

## *Certification*

**Graduate Aptitude test in Engineering 2020** (GATE score 460)

## *PhD Thesis*

***Title:*** Bis-Guanidinate-Supported Magnesium Complexes Synthesis, Reactivity, Catalytic Applications

***Supervisor:*** Prof. Sharanappa Nembenna

***Time Period:*** 2021-2026 (Ongoing)

***Key Research Contributions:***

- Development of bis-guanidinate magnesium (II) amide, and hydride complexes
- Reduction of carbodiimides and cyanamides using bis-guanidinate Mg hydride catalyst
- Substrate-assisted hydroboration of nitriles and isocyanides using Mg amide catalyst
- Development of bis-guanidinate magnesium chalcogenide chemistry (yet to publish)

## *Languages*

Bengali, Hindi, English

## *Conferences & Awards*

- *Main Group Molecules to Materials (MMM-IV)*, 2025 — Poster Presentation (**Best Poster Award**)
- *International Conference on Main Group Synthesis and Catalysis (ICMGSC-II)*, 2025 — Poster Presentation
- *International Conference on Main Group Synthesis and Catalysis (ICMGSC-I)*, 2023 — Poster Presentation
- *International Conference on Modern Trends in Inorganic Chemistry (MTIC-XIX)*, 2022 — Poster Presentation
- *Main Group Molecules to Materials (MMM-II)*, 2021 — Attended
- *DAE Conclave*, 2024 — Poster Presentation

## *Publications*

### Journal Articles

1. **Mukhopadhyay, S.**; Rajput, S.; Sahoo, R. K.; Nembenna, S., Insights Into the Magnesium-Catalyzed C–C Coupling of Terminal Alkynes with Carbodiimides. *Eur. J. Org. Chem.* **2024,** *27* (44), e202400702.
2. **Mukhopadhyay, S.**; Padhan, S.; Das, A.; Nembenna, S., Bis-guanidinate magnesium amide-catalyzed hydroboration of nitriles and isocyanides via a substrate-assisted pathway. *Org. Chem. Front.* **2026,** *13* (2), 350-363.
3. **Mukhopadhyay, S.**; Das, A.; Padhan, S.; Nembenna, S., Bulky bis-guanidinate ligand-supported magnesium hydride: Synthesis, reactivity, and catalytic applications. *Inorg. Chim. Acta.* **2026,** *589*, 122924.
4. **Mukhopadhyay, S.**; Das, A.; Rajput, S.; Behera, G.; Nembenna, S., Mechanistic Studies of Bis(Guanidinate) Magnesium Hydride as a Precatalyst for the Hydroboration of Organic Carbonates and Carbon Dioxide. *Catal. Sci. Technol.* (**2026**), 10.1039/D6CY00898D.
5. **Mukhopadhyay, S.**; Das, A.; Rajput, S.; Prajapati, D. J.; Nembenna, S., Reactivity of Bis(Guanidinate) Magnesium Hydride and Alkyl Complexes: Mg-E (E = S, Se, Te) Bond Formation and Structural Diversification. (*Inorg. Chem.* **2026,** 10.1021/acs.inorgchem.6c03268)
6. **Mukhopadhyay, S.**; Sahoo, R. K.; Patro, A. G.; Khuntia, A. P.; Nembenna, S., Low-valent germanium and tin hydrides as catalysts for hydroboration, hydrodeoxygenation (HDO), and hydrodesulfurization (HDS) of heterocumulenes. *Dalton Trans.* **2024,** *53* (45), 18207-18216.
7. Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.**; Nembenna, S., Organoaluminum Cation Catalyzed Selective Hydrosilylation of Carbonyls, Alkenes, and Alkynes. *Eur. J. Inorg. Chem.* **2022,** *2022* (8), e202101030.
8. Das, A.; **Mukhopadhyay, S.**; Rajput, S.; Nembenna, S., Magnesium catalyzed hydroamination of carbodiimides and hydroboration of cyanamides. *Dalton Trans.* **2025,** *54* (47), 17553-17563.
9. Sahoo, R. K.; **Mukhopadhyay, S.**; Rajput, S.; Jana, A.; Nembenna, S., Bulky bis-guanidine ligand-based neutral and cationic zinc alkyl, halide, and hydride complexes: synthesis, characterization, and catalytic application. *Dalton Trans.* **2025,** *54* (41), 15554-15560.

## *Publications*

10. Das, A.; **<u>Mukhopadhyay, S.</u>**; Nembenna, S., Conversion of Alkynes to Imidazolidin-2-ones: Insight into the Role of Magnesium Amido Complexes in the Reaction. *Organometallics* **2026,** *45* (1), 36-47.
11. Das, A.; **<u>Mukhopadhyay, S.</u>**; Rajput, S.; Nembenna, S., Catalytic hydroboration of α,β-unsaturated ketones using β-diketiminate magnesium hydride and mechanistic insights. *Dalton Trans.* **2026,** *55* (15), 6056-6066.
12. Padhan, S.; Rajput, S.; **<u>Mukhopadhyay, S.</u>**; Sabharwal, G.; Nembenna, S., Bis-Guanidinate Aluminum Complexes: Synthesis and Catalytic Application in Three-Component Aminophosphorylation of Aldehydes. *Organometallics* **2026,** *45* (12), 1384-1394.
13. Rajput, S.; Padhan, S.; Thilakan, N. M.; **Mukhopadhyay, S.**; Nembenna, S., Nickel-catalyzed dehydrogenative Zn–Zn coupling to a Zn(i) dimer and its reactivity. *Chem. Commun.* **2026,** *62* (2), 534-538.
14. Thilakan, N. M.; Rajput, S.; **<u>Mukhopadhyay, S.</u>**; Nembenna, S., Zinc-Mediated C–H Borylation of Terminal Alkynes: Mechanistic Insights. *Asian J. Org. Chem.* **2026,** *15* (3), e00637.

**Review Article**

15. **Mukhopadhyay, S.**; Patro, A. G.; Vadavi, R. S.; Nembenna, S., Coordination Chemistry of Main Group Metals with Organic Isocyanides. *Eur. J. Inorg. Chem.* **2022,** *2022* (31), e202200469.

**Book Chapters**

16. Nembenna, S.; Sarkar, N.; Sahoo, R. K.; **Mukhopadhyay, S.**, 2.03 - Organometallic Complexes of the Alkaline Earth Metals. In *Comprehensive Organometallic Chemistry IV*, Parkin, G.; Meyer, K.; O'hare, D., Eds. Elsevier: Oxford, 2022; pp 71-241.
17. **Mukhopadhyay, S.**; Rajput, S.; Nembenna, S., Guanidinate Ligand Stabilized Alkaline Earth Metal Complexes, Synthesis, Reactivity, and Catalytic Applications. In *Encycl. Inorg. Bioinorg. Chem.*, 2023; pp 1-21.
18. Prajapati, D. J.; Das, A.; **<u>Mukhopadhyay, S.</u>**; Nembenna, S., Zinc-Catalyzed Hydrofunctionalization Reactions. In *Encycl. Inorg. Bioinorg. Chem.*, 2026; pp 1-50.

## *Referee*

**Prof. Sharanappa Nembenna**
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